Critical points and Betti numbers in 2D random fields

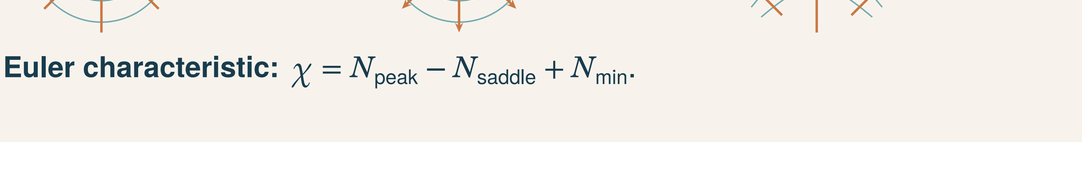
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Imagine a random field

Our goal: predict the geometric and topological statistics of 2D random fields.



The **threshold** ν acts like a water level: we count features above it.

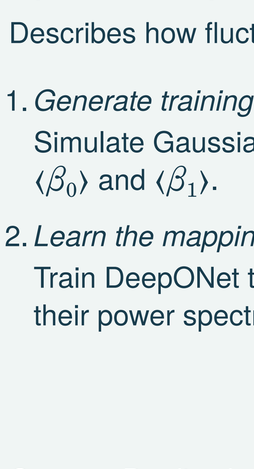
Local geometry: how many peaks, minima and saddles lie above ν?

Global topology: how many islands and holes does the land above ν form?

Toy model: $f = g + Ag^2 + Bg^3$, with a Gaussian field g.

Count islands and holes

As ν falls, **critical points change the topology**.



Peak
An island is born
$\Delta\beta_0 = +1$

Saddle: merge
Two islands join
$\Delta\beta_0 = -1$

Saddle: loop
A hole appears
$\Delta\beta_1 = +1$

Minimum
A hole fills
$\Delta\beta_1 = -1$

$$E_\nu = \{(x,y) : f(x,y) > \nu\}$$
$$\beta_i(\nu) = \dim H_i(E_\nu; \mathbb{F}_2)$$

β_0: connected components.

β_1: independent loops.

$\chi = \beta_0 - \beta_1$

Morse counts (planar case)

$$\beta_0(\nu) = N_{\text{peak}} - N_{\text{saddle,merge}},$$
$$\beta_1(\nu) = N_{\text{saddle,loop}} - N_{\text{min}}.$$

Count peaks, minima & saddles

Critical points: $\nabla f = 0$; arrows point uphill, along ∇f.



Euler characteristic: $\chi = N_{\text{peak}} - N_{\text{saddle}} + N_{\text{min}}$.

Can AI Predict Topology?

Input: Power spectrum $P(k)$

Describes how fluctuations are distributed across spatial scales.

1. *Generate training data*
   Simulate Gaussian fields with diverse $P(k)$ and measure the mean Betti curves $\langle\beta_0\rangle$ and $\langle\beta_1\rangle$.
2. *Learn the mapping*
   Train DeepONet to predict the topology of unseen Gaussian fields directly from their power spectra.

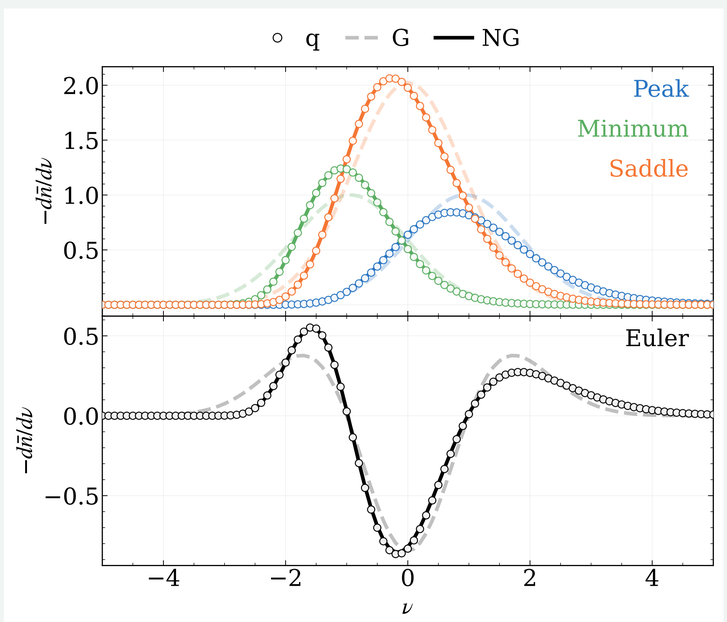
$$P(k) \xrightarrow{\text{DeepONet}} \{\langle\beta_0(\nu)\rangle, \langle\beta_1(\nu)\rangle\}$$

Output: Predict the average topology directly from $P(k)$, without running new simulations.

Statistical prediction

Kac–Rice formula

$$\rho_{\text{crit}}(\nu) = \left\langle \delta(F-\nu)\, \delta^{(2)}(\nabla F)\, |\det H_F| \right\rangle$$
$$F = (f - \langle f\rangle)/\sigma_f, \quad H_F = \nabla\nabla F$$



$\rho(\nu) = -d\bar{n}/d\nu$: cumulative critical-point number density

q: numerical measurements

G / NG: Predictions assuming Gaussian / non-Gaussian variable distributions

$A = 0.1, \quad B = 0.005.$

Monotonic Transformation of a Gaussian Field

$$\nabla f = f'(g)\nabla g, \qquad \nabla\nabla f = f'(g)\nabla\nabla g$$

with $f'(g) = 1 + 2Ag + 3Bg^2 > 0$.

⇒ Preserves critical points and types; changes only their heights.

Conserve the number of each type:

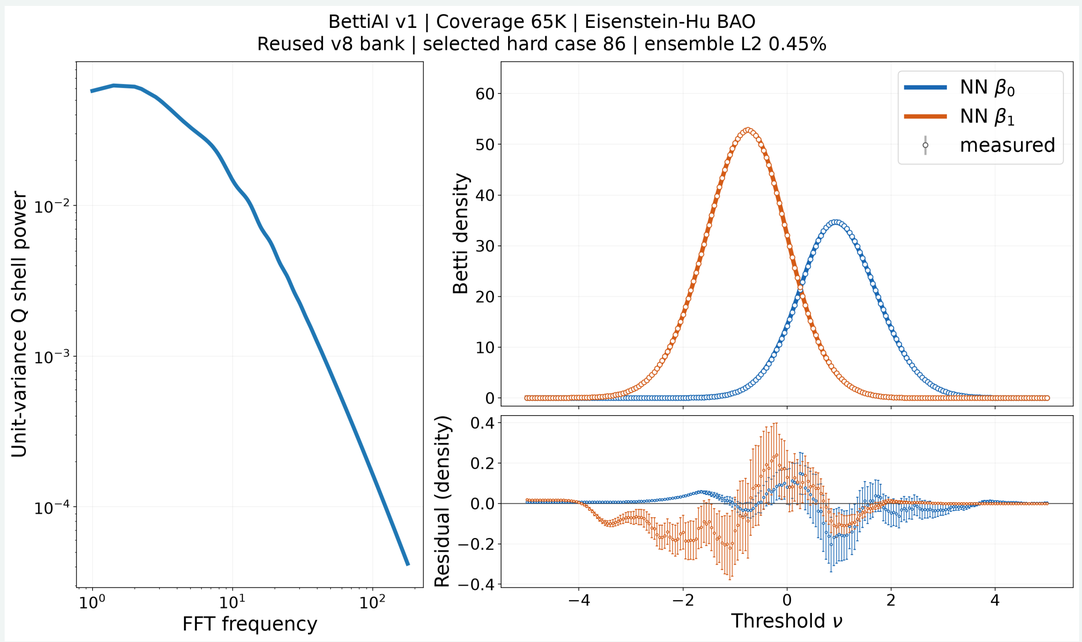
$$\rho_{\text{NG}}(\nu)\, d\nu = \rho_{\text{G}}(\nu_g)\, d\nu_g, \quad \Rightarrow \quad \rho_{\text{NG}}(\nu) = \rho_{\text{G}}(\nu_g)\, \frac{\sigma_f}{f'(\nu_g)}$$

ν_g is the inverse height map; σ_f is the standard deviation of f.

Monotonic transformations also preserve excursion-set topology:

$$\{f \geq \sigma_f \nu\} = \{g \geq \nu_g(\nu)\}, \quad \Rightarrow \quad \beta_i^{\text{NG}}(\nu) = \beta_i^{\text{G}}(\nu_g(\nu))$$

Topology is preserved; only thresholds are remapped.



The spectra still show several-percent errors. Generalization to spectral shapes poorly represented in the training set remains a key challenge.

Monotonic transformations extend these predictions through threshold remapping. More general non-Gaussian fields remain a future goal.

References & Acknowledgments

[1] T. Matsubara, Phys. Rev. D **101**, 043532 (2020).

[2] L. Lu et al., Nat. Mach. Intell. **3**, 218–229 (2021).

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